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SEMESTER 1, 2021 EXAMINATIONS**CITS2200****Physics, Mathematics & Computing****Data Structures and Algorithms**Department of Computer Science & Software
EngineeringThis paper contains: **5 Pages (including title page)**Time Allowed: **2:00** hours**INSTRUCTIONS:**

This paper has 5 pages and 5 questions.

Students should answer all questions.

Students may bring 1 page (2 sides) of handwritten notes for the exam. These notes are to be submitted with the exam answers.

THIS IS A CLOSED BOOK EXAMINATION (SEE ALLOWABLE ITEMS)**SUPPLIED STATIONERY**

1 x Answer booklet 10 pages

ALLOWABLE ITEMSUWA Approved Calculator with Sticker
Student Notes Only. Please Specify Below.**PLEASE NOTE**

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found. Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Candidates must comply with the Examination Rules of the University and with the directions of supervisors.

No electronic devices are permitted during the examination.

All question papers and answer booklets are the property of the University and remain so at all times.

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Q1.

(a)

Explain the merge step of the `mergesort` algorithm. What is the overall complexity of merging in the `mergesort` algorithm for an input of size n ? Explain your answer.

(6)

(b)

Explain the `heap` data structure. Explain how the `heap` data structure can be used for sorting an unsorted array of n integers in ascending order. What is the complexity of this algorithm? Explain clearly.

(6)

Q2.

(a)

Write a class definition including a constructor in Java for the `Link` class for implementing a singly linked list. Write a class definition including a constructor in Java for the `DoubleLink` class for implementing a doubly linked list. You can assume that the item stored in a node is an integer.

(6)

(b)

You are given a reference `current` of a node in a doubly linked list. Write a Java method `insertBefore()` that inserts another node before `current` with an item `e`. You can assume that `current` is neither the first node nor the last node in the list (that means you need not take care of special cases).

(6)

Q3.

(a)

Explain why

1. $O(n^2)$ is $O(n^3)$
2. $O(n^2)$ is not $O(n)$

(6)

(b)

Explain clearly when amortized complexity analysis is useful and important. Consider an operation `multiDequeue(i)` for a queue. `multiDequeue(i)` removes `i` elements from the front of a queue. What is the amortized complexity of `multiDequeue(i)`?

(6)

Q4.

(a)

Explain the inorder, preorder and postorder traversals of a tree. Explain how breadth-first search can be used for determining whether an undirected graph has one or more connected components.

(6)

(b)

Explain clearly Kruskal's minimum spanning tree algorithm. Explain how Kruskal's algorithm avoids adding cycles to the spanning tree. What is the complexity of Kruskal's algorithm? Explain your answer.

(6)

Q5.

(a)

Explain how Dijkstra's single-source-shortest-path algorithm can be implemented using a priority search tree. What is the complexity of Dijkstra's algorithm? Explain your answer.

(6)

(b)

Explain clearly how dynamic programming is used in the Floyd-Warshall algorithm (Note: Just writing the pseudo-code of the Floyd-Warshall algorithm without explanation is not an answer for this question). What is the complexity of the Floyd-Warshall algorithm? Explain your answer.

(6)

-END OF PAPER-
